## 4.3 Space with Functions

**Definition 4.3.1 (Space with functions)**
*Let $X$ be a topological space and $\mathcal{O}_X$ be a subsheaf of the $\overline{k}$-valued functions on $X$ (which is a sheaf of $k$-algebras). Then we say $(X, \mathcal{O}_X)$ is a **space with functions**. Additionally it's **local** if it satisfies the condition*

*a) For every $f \in \mathcal{O}_X(U)$ we have $D(f) \coloneqq \{f \neq 0\}$ is open and $\frac{1}{f} \in \mathcal{O}_X(D(f))$.*

*Note that the sheaf restriction maps $\rho_{UV} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ correspond simply to restrictions of functions.*

*In the latter case $X$ is a locally ringed space over $k$ because $\mathcal{O}_{X,x}$ is a local $k$-algebra with maximal ideal $\mathfrak{m}_{X,x}$ consisting of stalks vanishing at $x$.*

*A **regular morphism** or **regular map** $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous function $\phi : X \to Y$ such that*

$$g \in \mathcal{O}_Y(U) \implies g \circ \phi \in \mathcal{O}_X\big(\phi^{-1}(U)\big)$$

*This determines a morphism of sheaves*

$$\phi^\sharp : \mathcal{O}_Y \longrightarrow \phi_\star \mathcal{O}_X$$

*We say a regular morphism is **dominant** if $\phi(X)$ is dense in $Y$.*

*An isomorphism is a homeomorphism and induces an isomorphism of sheaves, or equivalently has a two-sided regular inverse.*

**Proposition 4.3.2**
*The composition of regular maps is well-defined and associative.*

### 4.3.1 Local Rings

**Definition 4.3.3 (Local Ring)**
*Let $X$ be a local space with functions and $W \subset X$ an irreducible closed subset. Then we define the **local ring** at $W$ to be (see (4.2.5))*

$$\mathcal{O}_{X,W} \coloneqq \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

*If $\overline{W} = \overline{W'}$ then $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$, so for example $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$.*

*It is a local ring with unique maximal ideal*

$$\mathfrak{m}_{X,W} \coloneqq \{[(V,\sigma)] \mid V \cap W \neq \emptyset, \sigma \in \mathcal{O}_X(V), \sigma(x) = 0 \; \forall x \in V \cap W\}$$

*Define the **residue field***

$$k\big(\mathfrak{m}_{X,W}\big) \coloneqq \mathcal{O}_{X,W}/\mathfrak{m}_{X,W}$$

*In the case $W = \{x\}$ we write $\mathcal{O}_{X,x}$, $\mathfrak{m}_{X,x}$ and $k\big(\mathfrak{m}_{X,x}\big)$.*

*Proof.* Consider $(V,\sigma) \in \mathcal{O}_{X,W} \setminus \mathfrak{m}_{X,W}$. Then by definition $D(\sigma) \cap W \neq \emptyset$ and $(D(\sigma), \sigma^{-1})$ is an explicit inverse. ∎

**Corollary 4.3.4 (Field of Rational Functions)**
*Let $X$ be an irreducible space with functions. Then $\mathcal{O}_{X,X} =: k(X)$ is a field, which we call the **field of rational functions**.*

**Proposition 4.3.5 (Stalk Maps)**
*Suppose $\phi : X \to Y$ is a regular morphism and $Z \subset \overline{\phi(W)}$. Then there exists a unique (local) $k$-algebra homomorphism $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ such that the following diagram commutes*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\rho \circ \phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

*for all open sets $V \subset Y$ for which $V \cap Z \neq \emptyset$ and some $U \subset \phi^{-1}(V)$ for which $U \cap W \neq \emptyset$. This is functorial in the sense that*

© David Rufino CC BY-NC-ND 4.0

$$(\psi \circ \phi)_W = \psi_Z \circ \phi_W$$

*Proof.* If $V$ meets $Z$ then by (4.1.19) $\phi^{-1}(V)$ meets $W$. Therefore the vertical maps are well-defined. We define

$$\phi_W((V', \sigma)) := (\phi^{-1}(V'), \sigma \circ \phi)$$

and immediately verify it is well-defined and ensures the diagram commutes. ∎

**Proposition 4.3.6 (Stalk Maps II)**
*Let $\phi : X \to Y$ be a regular morphism of local spaces with functions. Suppose $y \in \overline{\{\phi(x)\}}$. Then there is a unique local algebra homomorphism $\phi_x : \mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$ making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \dashrightarrow & \mathcal{O}_{X,x} \end{array}$$

*for all open neighbourhoods $V$ of $\phi(x)$ and some open neighbourhood $U \subset \phi^{-1}(V)$ of $x$. This is functorial in the sense that*

$$(\psi \circ \phi)_x = \psi_{\phi(x)} \circ \phi_x$$

*Proof.* This follows immediately from (4.3.5). ∎

**Proposition 4.3.7 (Stalk Maps Function Field)**
*Let $X, Y$ be irreducible local spaces with functions and $\phi : X \to Y$ a dominant regular morphism. Then there is a unique homomorphism $\phi_\star : k(Y) \hookrightarrow k(X)$ making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ k(Y) & \overset{\phi_\star}{\hookrightarrow} & k(X) \end{array}$$

*for all non-empty open subsets $V \subset Y$ and some non-empty open subset $U \subset \phi^{-1}(V)$. This is functorial in the sense that*

$$(\phi \circ \psi)_\star = \psi_\star \circ \phi_\star$$

*Proof.* This follows immediately from (4.3.5) with $W = X$ and $T = Y$. ∎

**Proposition 4.3.8**
*Let $X$ be a local space with functions and $Z, W$ be irreducible subsets such that $Z \subset \overline{W}$. Then there is a canonical homomorphism $\mathcal{O}_{X,Z} \to \mathcal{O}_{X,W}$ making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_X(U) & & \\ \downarrow & \searrow & \\ \mathcal{O}_{X,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

*for all open subsets $U$ of $X$ meeting $Z$.*

*This holds in particular when $X$ is irreducible we may take $W = X$ and $\mathcal{O}_{X,W} = k(X)$.*

*Proof.* This follows immediately from (4.3.5) by considering the case $\phi = 1_X$. ∎

**Proposition 4.3.9 (Stalk Maps under Specialisation)**
*Let $X, Y$ be irreducible local spaces with functions, $\phi : X \to Y$ a dominant regular morphism and $x \in X$ such that $y \in \overline{\{\phi(x)\}}$. Then for all neighbourhoods $V$ of $\phi(x)$ we have a commutative diagram*

© David Rufino CC BY-NC-ND 4.0

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,y} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xrightarrow{\phi_\star} & k(X) \end{array}$$

**Lemma 4.3.10 (Evaluation of Stalks)**
*Let $X$ be a local space of functions and $x \in X$. Then there is a unique function*

$$\mathrm{ev}_x : k\big(\mathfrak{m}_{X,x}\big) \hookrightarrow \overline{k}$$

*making the following diagram commute for all open neighbourhoods $U$ of $x$*

$$\begin{array}{ccccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_{X,x} & \longrightarrow & k\big(\mathfrak{m}_{X,x}\big) \\ & \searrow^{\mathrm{ev}_x} & & & \downarrow \mathrm{ev}_x \\ & & & & \overline{k} \end{array}$$

*This is functorial in the sense that for a regular morphism $\phi : X \to Y$ there is a commutative diagram*

$$\begin{array}{ccc} k\big(\mathfrak{m}_{Y,\phi(x)}\big) & \xrightarrow{\phi_x} & k\big(\mathfrak{m}_{X,x}\big) \\ \downarrow \mathrm{ev}_{\phi(x)} & & \downarrow \mathrm{ev}_x \\ \overline{k} & = & \overline{k} \end{array}$$

## 4.3.2 Open Immersions

**Definition 4.3.11 (Open Immersion)**
*Let $X, Y$ be spaces with functions. Then we say a regular map $\phi : X \to Y$ is an **open immersion** if*

*a) $\phi$ is an open embedding, and*
*b) $\phi$ induces an isomorphism of sheaves*

$$\phi_\star\big(\mathcal{O}_Y|_{\phi(X)}\big) \xrightarrow{\sim} \mathcal{O}_X$$

**Proposition 4.3.12 (Open Subspace)**
*Let $X$ be a (local) space of functions and $U \subset X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a (local) space of functions which we call an **open subspace**. We may write $\mathcal{O}_U \coloneqq \mathcal{O}_X|_U$.*

*The inclusion $i : U \hookrightarrow X$ is an open immersion and an open immersion $\phi : X \to Y$ induces an isomorphism between $X$ and the open subvariety $\phi(X)$.*

**Proposition 4.3.13**
*Let $X$ be a space of functions and $V \subset U$ open subsets. Then $(V, \mathcal{O}_U|_V) = (V, \mathcal{O}_X|_V)$.*

**Proposition 4.3.14**
*Let $\phi : X \to Y$ be an open embedding and $Z \subset Y$ closed. Then $Z$ is irreducible iff $\phi^{-1}(Z)$ is irreducible.*

*In particular if $Y$ is irreducible then so is $X$.*

*Proof.* We may reduce to the case $X = U$ is an open subset of $Y$. Then it follows from (4.1.73) and (4.1.70). ∎

**Proposition 4.3.15**
*Let $\phi : X \to Y$ be a regular open embedding. Then the following are equivalent*

*a) $\phi$ is an open immersion*
*b) for every irreducible subset $W \subset X$ the canonical map*

$$\mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$$

© David Rufino CC BY-NC-ND 4.0

*is an isomorphism for all $Z$ such that $\overline{Z} = \overline{\phi(W)}$*

*c) b) holds for some $Z$ such that $\overline{Z} = \overline{\phi(W)}$*

*In particular for all open subsets $V \subset Y$ and $U \subset \phi^{-1}(V)$ such that $U \cap W \neq \emptyset$ we have a commutative diagram*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(W)} & \xrightarrow{\sim} & \mathcal{O}_{X,W} \end{array}$$

*Proof.* $b) \implies a)$ By definition we may factor into a regular homeomorphism $X \xrightarrow{\sim} V \subset Y$. It is an isomorphism of spaces by (4.2.11).

$a) \implies b)$ We may reduce to the case that $X \subset Y$ is an open subset. Because an open subset of $X$ intersects $W$ if and only if it intersects $Z$ we find that the map is bijective.

$c) \iff b)$ we simply observe that if $\overline{Z} = \overline{Z'}$ then the canonical maps are the same. ∎

**Proposition 4.3.16**
*Let $\phi : X \to Y$ be a dominant open immersion. If $Y$ is irreducible, then so is $X$ and the map*

$$\phi_\star : k(Y) \hookrightarrow k(X)$$

*is an isomorphism.*

*In particular if $X$ is irreducible and $U \subset X$ is a non-empty open subset, then there is a canonical isomorphism*

$$k(X) \xrightarrow{\sim} k(U)$$

*Proof.* By (4.3.14) $X$ is irreducible, so it is a special case of (4.3.15). ∎

**Proposition 4.3.17 (Stalk Maps under Open Immersions I)**
*Let $X$ be a local space with functions, $U \subset X$ an open subset and $W \subset X$ a irreducible closed subset such that $U \cap W \neq \emptyset$. Then $U \cap W$ is an irreducible subset of $U$ and we have canonical isomorphisms of stalks making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_X(V) & \longrightarrow & \mathcal{O}_U(U \cap V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{\sim} & \mathcal{O}_{U,W\cap U} \end{array}$$

*for all open $V \subset X$ for which $V \cap W \neq \emptyset$.*

*Proof.* The inclusion $i : U \hookrightarrow X$ is an open immersion, so $U \cap W$ is irreducible by (4.1.73). This is then a special case of (4.3.15). ∎

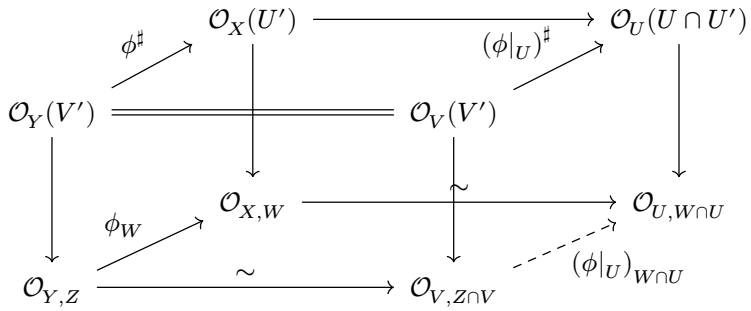
**Lemma 4.3.18**
*Let $U, Z$ be subsets of a topological space $X$, with $U$ open. Then*

$$\overline{Z} \cap U \subset \overline{Z \cap U}$$

*Proof.* Suppose $x \in \overline{Z} \cap U$ and $V$ is an open subset containing $x$. Then $U \cap V$ is an open set containing $x$ and therefore by (4.1.19) $Z \cap U \cap V \neq \emptyset$. As $V$ was an arbitrary neighbourhood of $x$ we conclude that $x \in \overline{Z \cap U}$. ∎

**Proposition 4.3.19 (Restricting Stalk Maps to Open Subspaces)**
*Let $\phi : X \to Y$ be a regular morphism of local space with functions. Suppose $U \subset X$ and $V \subset Y$ are open subsets such that $U \subset \phi^{-1}(V)$, and $W \subset X$, $Z \subset Y$ are irreducible subsets such that $Z \subset \overline{\phi(W)}$ and $U \cap W \neq \emptyset$. Then $\phi(W \cap U) \subset \overline{Z \cap V}$ and there is a commutative cube for every open set $V' \subset V$ such that $V' \cap Z \neq \emptyset$ and $U' \subset \phi^{-1}(V')$ such that $U' \cap W \neq \emptyset$*

© David Rufino CC BY-NC-ND 4.0




*where the stalk isomorphisms are defined in (4.3.17) and $(\phi|_U)_{W\cap U}$ is the unique (local) homomorphism making the bottom square (or right side) commute*

*Proof.* Observe by (4.3.18) that

$$\phi(W \cap U) \subset \phi(W) \cap V \subset \overline{Z} \cap V \subset \overline{Z \cap V}$$

which means the stalk map of $\phi|_U$ is well-defined. The sides of the cube are well-defined and commute by (4.3.17) and (4.3.5), and the top by definition. As the horizontal arrows are isomorphisms then we see that the bottom square commutes, and $(\phi|_U)_{W\cap U}$ is the unique homomorphism for which this is the case. ∎

### 4.3.3 Closed Immersions

**Definition 4.3.20 (Closed Immersion)**
*Let $\phi : X \to Y$ be a regular morphism of spaces with functions. We say that $\phi$ is a **closed immersion** if*

*a) $\phi$ is a homeomorphism onto a closed subset $\phi(X)$ of $Y$*
*b) For all $x \in X$ we have the induced map on stalks*

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_X$$

*is surjective.*

**Proposition 4.3.21 (Closed Subset Structure Sheaf)**
*Let $(X, \mathcal{O}_X)$ be a space of functions and $Z \subset X$ a closed subset. Then there exists a unique sheaf of functions $\mathcal{O}_Z$ such that the inclusion $j : Z \to X$ is a closed immersion.*

*More precisely for $U \subset Z$ an open subset*

$$\mathcal{O}_Z(U) := \left\{ f : U \to \overline{k} \;\middle|\; f(y) = g_i(y)\ \forall y \in Z \cap U_i \text{ for some } g_i \in \mathcal{O}_X(U_i) \text{ where } Z \cap \bigcup_{i\in I} U_i = U \right\}$$

*Proof.* Evidently $j$ is a homeomorphism onto $Z$. We first show that $j$ is regular. Suppose $g \in \mathcal{O}_X(U')$ then by definition we have $g \circ j \in \mathcal{O}_Z(j^{-1}(U')) = \mathcal{O}_Z(U' \cap Z)$.

Given $(V, f) \in \mathcal{O}_{Z,z}$ by taking a smaller open subset we may suppose that $f(y) = g(y)$ for all $y \in V$ for some $g \in \mathcal{O}_X(U)$ and $U \cap Z = V$. Then the image of $(U, g)$ under the stalk map is $(U \cap Z, g \circ j)$ which is precisely $(V, f)$. Therefore the stalk map $j_z$ is surjective for all $z \in Z$.

Let $\mathcal{F}$ be a sheaf of functions for which $j$ is a (regular) closed immersion and suppose $f \in \mathcal{O}_Z(U)$. Then $f|_{U_i\cap Z}(y) = g_i(y)$ for all $y \in Z \cap U_i$ and $g_i \in \mathcal{O}_X(U_i)$. By assumption $f|_{U_i\cap Z} = g_i \circ j \in \mathcal{F}(U_i \cap Z)$. By the sheaf condition we have $f \in \mathcal{F}(U)$ which shows $\mathcal{O}_Z$ is a subsheaf of $\mathcal{F}$.

We need to show that the inclusion of sheaves $\mathcal{O}_Z \hookrightarrow \mathcal{F}$ is an isomorphism. By (4.2.28) c) $\implies$ d) and (4.2.11) it is enough to show that it is surjective on stalks. However this follows by assumption by observing the composition $\mathcal{O}_{X,j(z)} \to \mathcal{O}_{Z,z} \to \mathcal{F}_z$ is surjective. ∎

**Remark 4.3.22**
*This can be defined formally as $j^{-1}(\mathcal{O}_X/\mathcal{I})$ where $\mathcal{I}$ is the presheaf of regular functions which vanish on $Z$. We show in the Algebraic Geometry chapter's treatment of closed subvarieties that this is consistent with the concept of affine subvariety.*

**Proposition 4.3.23**
*Let $\phi : X \to Y$ be a regular morphism of spaces with functions. Then the following are equivalent*

*a) $\phi$ is a closed immersion*

© David Rufino CC BY-NC-ND 4.0

*b) $\phi(X)$ is closed and $\phi$ determines an isomorphism $X \xrightarrow{\sim} \phi(X)$ of spaces with functions using the structure defined in (4.3.21)*

*Proof.* *a)* $\implies$ *b)* By definition $\phi(X)$ is closed and $\phi$ determines a homeomorphism onto $\phi(X)$. Let $Z \coloneqq \phi(X)$ and $\mathcal{O}_Z$ denote the canonical structure sheaf. There is a well-defined map

$$\mathcal{O}_Z(U) \to \mathcal{O}_X(\phi^{-1}(U))$$

which is injective, and therefore injective on stalks (4.2.28) c) $\implies$ d). By (4.2.11) we need only show that the map $\mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_{X,x}$ is surjective. However by hypothesis the composite map

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_x$$

is surjective from which the result follows.

*b)* $\implies$ *a)* Evidently $\phi$ is a homeomorphism onto $\phi(X)$. Further we have a sequence of stalk maps

$$\mathcal{O}_{Y,\phi(x)} \xrightarrow{\sim} \mathcal{O}_{\phi(X),\phi(x)} \to \mathcal{O}_{X,x}$$

where the first is surjective by (4.3.21) and the second is an isomorphism by assumption. Therefore the composite is surjective and $\phi$ is a closed immersion. ∎

## 4.3.4 Locally Closed Subspace

**Lemma 4.3.24 (Locally Closed Structure Sheaf)**
*Let $(X, \mathcal{O}_X)$ be a space with functions, $(Z, \mathcal{O}_Z)$ a closed subset with canonical structure defined in (4.3.21) and $(U, \mathcal{O}_U)$ an open subset with canonical structure (4.3.12). Then*

$$\mathcal{O}_Z|_{Z\cap U} = \mathcal{O}_U|_{Z\cap U}$$

## 4.3.5 Glueing

**Proposition 4.3.25 (Patching Topologies)**
*Let $X = \bigcup_{i=1}^n U_i$ be a set with topologies given for each $U_i$. Suppose that for all $i, j = 1...n$ the following properties hold*

- *$U_i \cap U_j$ is open in $U_i$, and*
- *the identity map $1 : U_i \cap U_j \to U_i \cap U_j$ is continuous with respect to the subspace topologies on $U_i$ and $U_j$ respectively.*

*Then there is a unique topology on $X$ such that*

- *the $U_i$ are open in $X$,*
- *the topology on $U_i$ coincides with the subspace topology from $X$*

*More precisely it is given by*

$$\{U \subset X \mid U \cap U_i \text{ open in } U_i \text{ for all } i = 1...n\}$$

**Proposition 4.3.26 (Glueing Topologies)**
*Let $X = \bigcup_{i=1}^n U_i$ be a set together with bijections $\theta_i : U_i \xrightarrow{\sim} W_i$ where $W_i$ are topological spaces. Suppose that for all $i, j = 1...n$ the following properties hold*

- *$\theta_i(U_i \cap U_j)$ is open in $W_i$,*
- *the map $\theta_{ij} \coloneqq \theta_j \circ \theta_i^{-1} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$ is continuous.*

*Then there is a unique topology on $X$ such that*

- *$U_i$ are open in $X$,*
- *$\theta_i$ is a homeomorphism with respect to the subspace topology on $U_i$ for all $i = 1...n$.*

*More precisely the topology is given by*

$$\{U \subset X \mid \theta_i(U \cap U_i) \text{ is open in } W_i \text{ for all } i = 1...n\}$$

**Proposition 4.3.27 (Glueing Regular Maps)**
*Let $X, Y$ be spaces with functions and $X = \bigcup_{i=1}^n U_i$ an open cover. Suppose there exists regular maps*

© David Rufino CC BY-NC-ND 4.0

$$\phi_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \to (Y, \mathcal{O}_Y)$$

*which satisfy* $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ *for all* $i, j = 1...n$*. Then there exists a unique map* $\phi : X \to Y$ *such that* $\phi|_{U_i} = \phi_i$*, which is regular.*

*Proof.* As a function $\phi$ is clearly well-defined and unique. It is continuous by (4.1.29). Suppose $g \in \mathcal{O}_Y(V)$ then by definition $(g \circ \phi)|_{\phi^{-1}(V) \cap U_i} = g \circ \phi_i \in \mathcal{O}_{U_i}(\phi^{-1}(V) \cap U_i) = \mathcal{O}_X(\phi^{-1}(V) \cap U_i)$. As $\mathcal{O}_X$ is a sheaf this shows that $g \circ \phi$ is regular as required. ∎

© David Rufino CC BY-NC-ND 4.0